



EcoSpray

AN IOT-BASED SPRINKLER AND MONITORING SYSTEM FOR PANABO PARK NURSERY

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INTRODUCTION

Traditional irrigation at Panabo Park Nursery relies heavily on manual watering, resulting in inconsistent irrigation schedules, water wastage, and increased labor requirements. To address these challenges, the researchers developed EcoSpray, an IoT-based sprinkler and monitoring system that automates irrigation using real-time environmental data.

OBJECTIVES

- Automate irrigation using soil moisture sensors
- Monitor temperature and humidity in real time
- Provide remote monitoring through a mobile application
- Reduce water consumption and manual labor
- Improve irrigation efficiency for ornamental plants

PROBLEM STATEMENT

The nursery experienced:

- Manual and inconsistent watering
- Water wastage
- Lack of real-time monitoring
- Difficulty managing irrigation schedules



METHODOLOGY

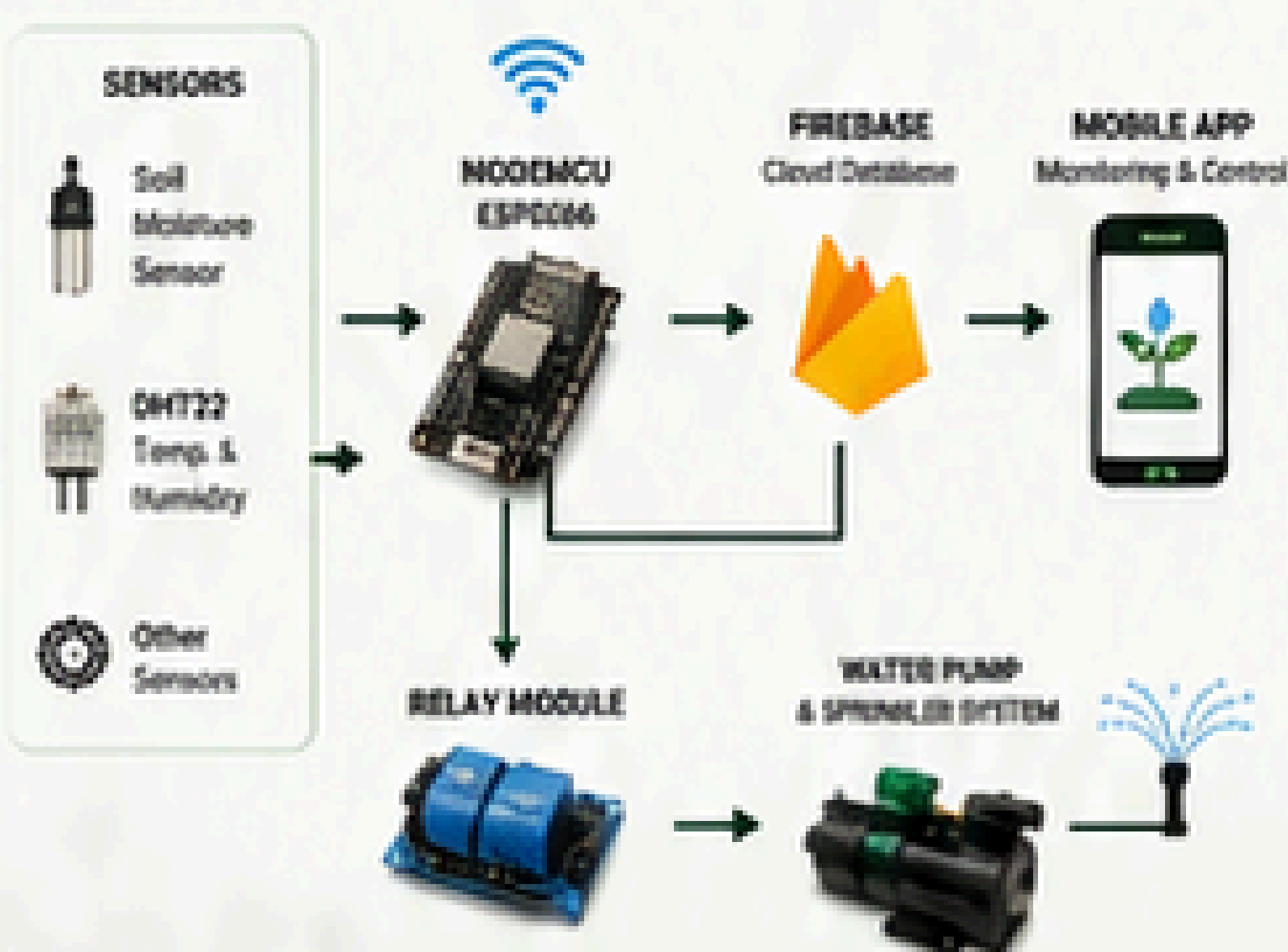
Development Method

The study utilized the Agile Software Development Life Cycle (SDLC).

- Planning
- Analysis
- Design
- Testing
- Implementation



SYSTEM ARCHITECTURE



Threshold Formula

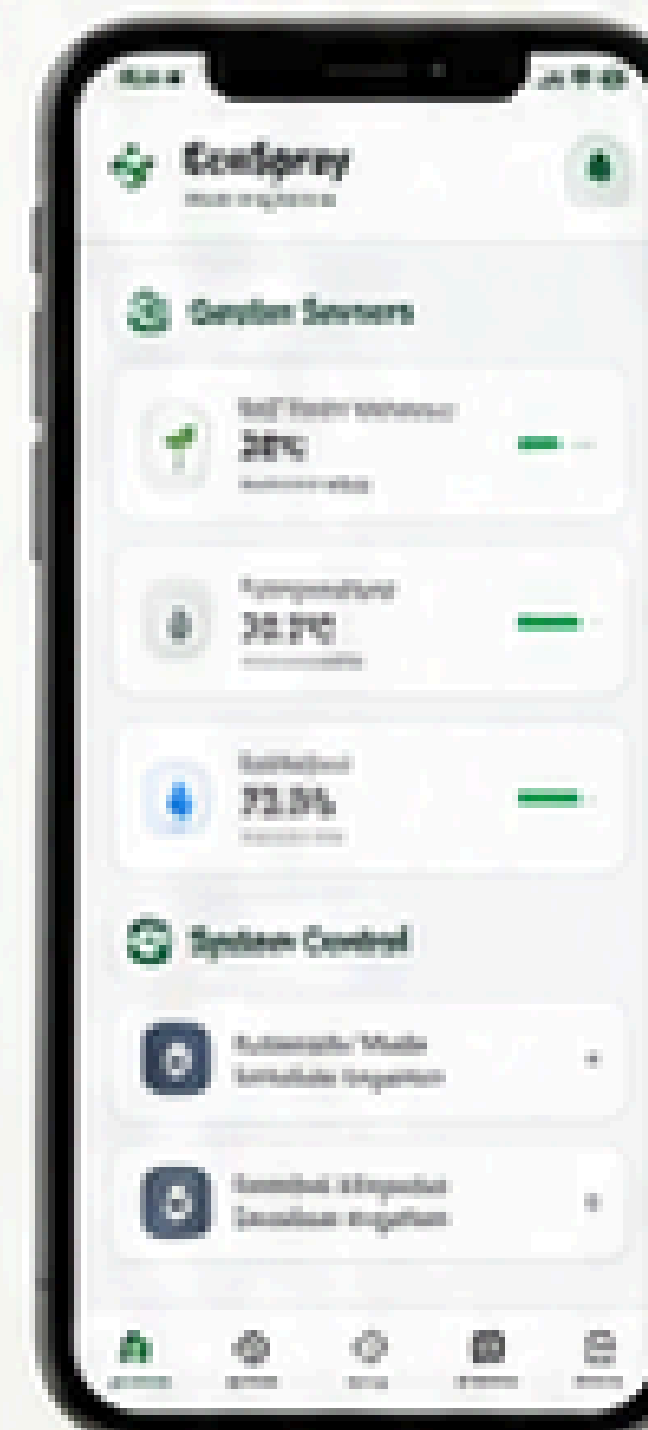
The system activates irrigation when soil moisture falls below:

Irrigation Activated if Soil Moisture < 35%

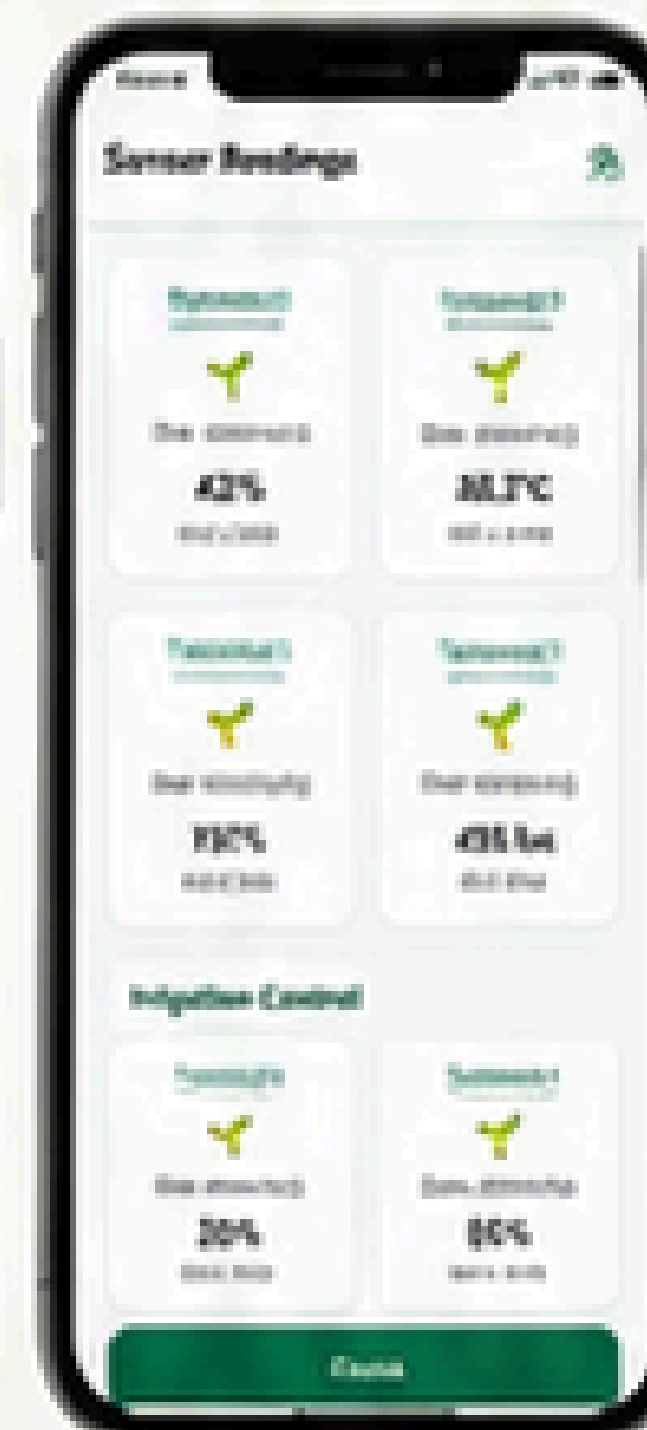
SYSTEM PROCESS

- Sensors collect environmental data
- NodeMCU processes sensor readings
- Data is sent to Firebase
- System activates sprinkler automatically when soil moisture drops below threshold
- Users monitor and control irrigation through mobile app

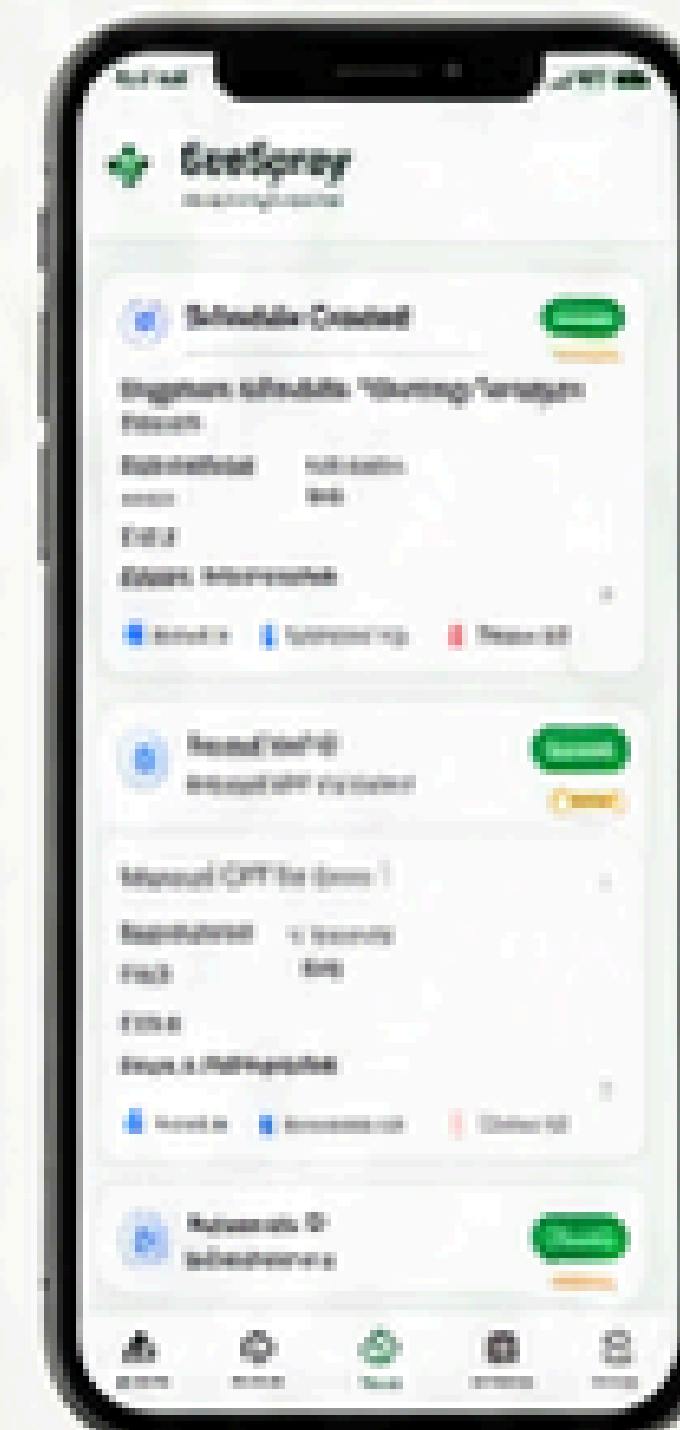
SYSTEM INTERFACE (MOBILE APP)



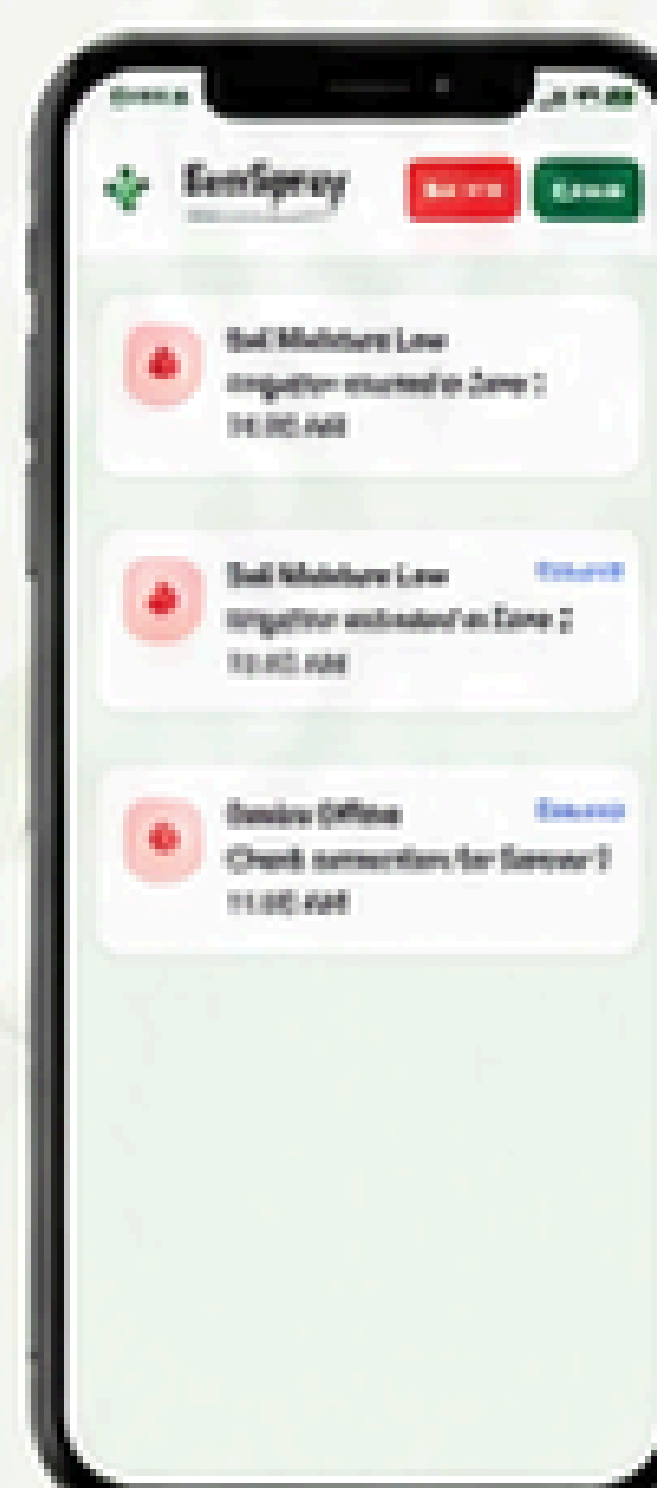
Dashboard



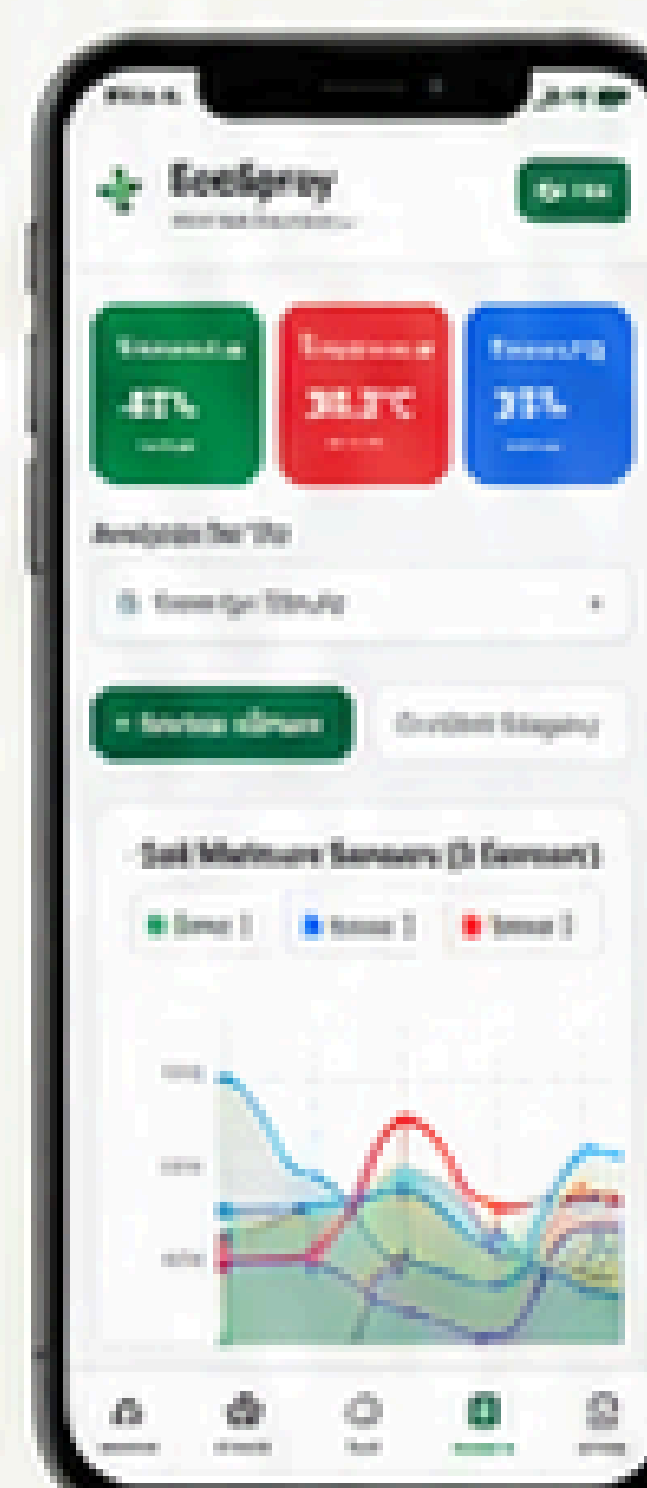
Control Panel



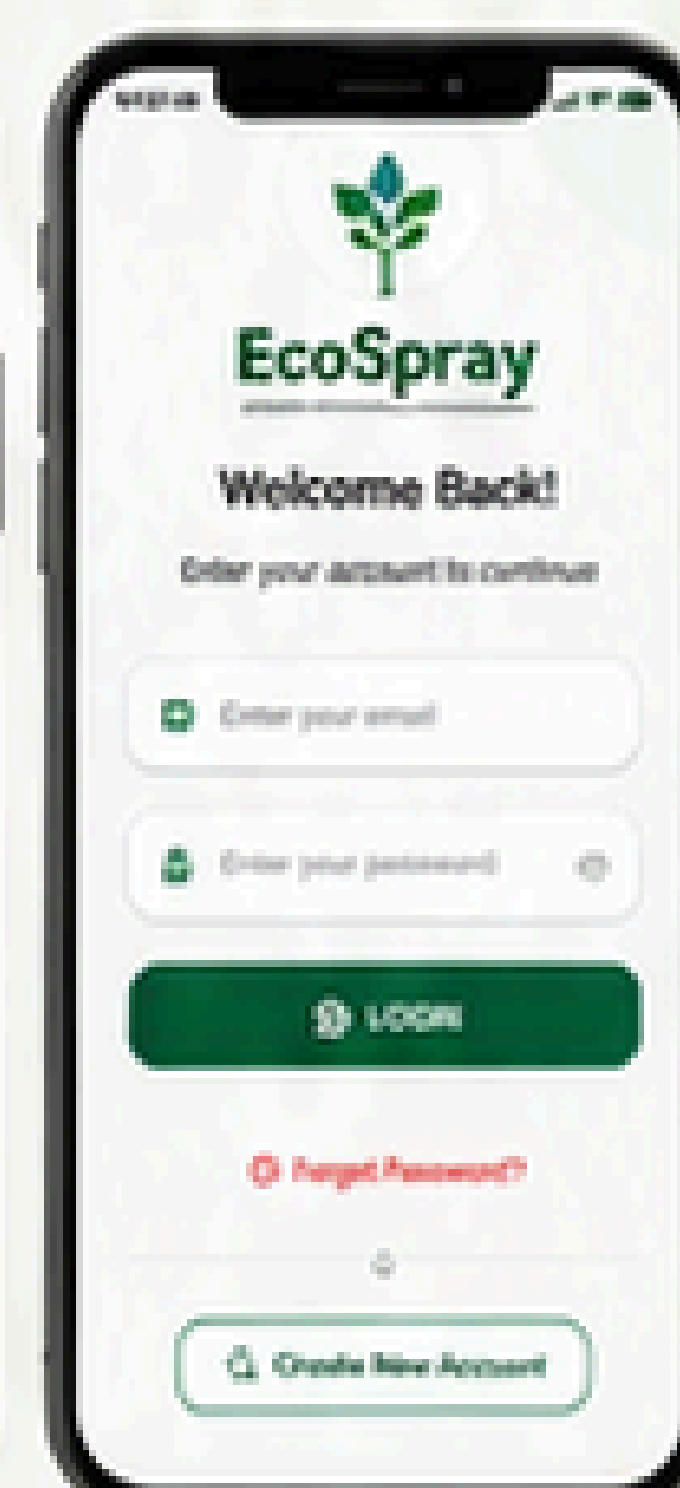
Activity Logs



Notifications



Analytics



Login Screen

RESULTS AND DISCUSSION

Key Findings

- Automated irrigation reduced manual watering by 70%
- Real-time monitoring improved irrigation management
- Mobile application enabled remote control and notifications
- Water distribution improved after pump enhancement
- Nursery staff observed increased operational efficiency

System Features

- Real-time Dashboard Monitoring
- Automatic Irrigation Scheduling
- Activity Logs and Analytics
- Mobile Notifications
- Manual and Automatic Control Modes

System Architecture (Implemented)



Issues Encountered

- Water pressure problem
- Low-pressure driplines
- Uneven water distribution

Solutions Implemented

- Repositioned hoses
- Improved sprinkler configuration
- Installed high-voltage water pump



Overall Outcome

The EcoSpray system successfully improved irrigation efficiency, reduced water waste, and supported sustainable plant care at Panabo Park Nursery.

BENEFITS

- Water Conservation**
Significantly reduces water waste
- Time & Labor Savings**
Automates irrigation tasks
- Healthier Plants**
Ensures consistent watering
- Data-Driven Decisions**
Real-time data for better planning
- Sustainable Irrigation**
Promotes eco-friendly nursery practices

KEY COMPONENTS



NodeMCU ESP8266 | Soil Moisture Sensor | DHT22 Temp. & Humidity Sensor | Relay Module | I2C 16x2 LCD | I2C 20x4 LCD | LED Lights (Red & Blue) | Water Pump (Booster Jet Pump) | 360° Nozzle Micro Sprayer Pump | PVC Water Hose | Mini Water Pump

FUTURE WORK

- Expand the system to more irrigation zones
- Integrate rainfall sensor for smarter irrigation
- Add solar power for off-grid operation
- Implement advanced analytics and AI prediction

Towards Smarter Irrigation,
Greener Tomorrow.

Save Water

Save Plants

Save Future